XP

KAIZEN XP

The RCFA Template

Root cause analysis for the faults
that keep coming back.

8 STEPS. ONE ROOT CAUSE. NO REPEATS.

FILLABLE ON SCREEN. PRINTABLE ON PAPER.

*"Do what nobody else can be bothered to do,
and you'll get what nobody else gets."*

FOR TECHNICIANS, ENGINEERS AND SHIFT LEADERS

READ THIS FIRST

Fix the symptom and you will be back here next week. Fix the cause and you never see it again.

RCFA stands for Root Cause Failure Analysis. It sounds like an engineering department thing. It is not. It is a structured way of asking "why did that really happen?" and refusing to stop at the first easy answer.

Most faults get fixed the same way every time: someone resets it, clears it, swaps the part, and the line runs again. Job done, until it happens on the next shift. The fault was never fixed. It was just survived.

This template walks you through eight steps, in order. Do them properly once and the fault stops coming back. Do that a few times and people start bringing their problems to you.

THE EIGHT STEPS

1. **Define the problem**
   Exactly what, where, when and how often. Facts, not opinions.
2. **Measure the impact**
   Downtime, waste, cost, safety, quality. This is why anyone should care.
3. **Contain it**
   Stop it hurting today, while you find the real cause.
4. **Build the timeline**
   What happened, in order, and how you know.
5. **List every possible cause**
   Use the six categories, then rule them out one by one with evidence.
6. **Find the root cause**
   5 Whys on what is left. Then prove it.
7. **Fix it for good**
   Actions with an owner and a date, aimed at the root cause.
8. **Check, lock it in, spread it**
   Prove it worked, make it the standard, and use it everywhere it applies.

THREE RULES

One problem per form. If you find two, start a second form.
If your root cause is a person ("operator error"), you have not gone deep enough. Ask why the system let it happen.
Containment is not a fix. A cone round a leak stops someone slipping. It does not stop the leak.

HOW TO USE THIS PDF

On screen: open it in the free Adobe Acrobat Reader app (phone or computer), or in Chrome or Edge. Tap a box to type. Boxes with a small gold arrow open a list to pick from. Save a copy each time you start a new sheet. Some built-in phone PDF viewers do not show the lists, so use Acrobat Reader if they are missing.
On paper: print the pages you need and write straight in the boxes. The lists work as write-in boxes.

WORKED EXAMPLE: LINE 3 LABELLER

One page summary of a real-world style RCFA, so you can see what good looks like at every step.

1. DEFINE	*Line 3 labeller jams at reel changes on all shifts, worst on nights. About 4 jams a shift since the new label stock started three weeks ago.*
2. IMPACT	*4 jams x 6 mins x 15 shifts = 6 hours downtime a week. About 40 mislabelled packs scrapped a week. No safety risk found.*
3. CONTAIN	*Operator checks every splice before restarting. Spare reels moved next to the machine to speed up recovery.*
4. TIMELINE	*Jams always within 2 minutes of a reel change. Started the week new label stock arrived. Fewer on days, when one experienced operator does most changes.*
5. CAUSES	*Ruled out: worn guide roller (measured, in spec), label stock width (measured, in spec), sensor (tested, fine). Confirmed: splice tape overhanging the web edge.*
6. ROOT CAUSE	*No standard method for splicing, so tape goes on at different angles. New stock is thinner, so overhang now catches. Verified: 20 splices done one way, zero jams.*
7. FIX	*Splice guide marked on the table. One-point lesson with photos. Every crew shown in person. Splicing added to the new starter checklist.*
8. CHECK & SPREAD	*Two weeks later: 1 jam a shift, down from 4. Splice guide added to Lines 1 and 2, which use the same labeller.*

WHAT TO NOTICE

The first theory, a worn roller, would have cost a part and fixed nothing. Every cause was ruled in or out with a measurement, and the root cause was proven before anyone spent a penny.

STEP 1: DEFINE THE PROBLEM

If you cannot describe it exactly, you cannot fix it exactly.

REF | DATE RAISED | LINE / EQUIPMENT

LEAD | TEAM

THE FACTS

Write what you saw or measured. Leave opinions for later.

WHAT happened?	
WHERE exactly?	
WHEN did it start?	
WHO found it?	
WHICH product, part or batch?	
HOW does it show up?	
HOW MUCH, and HOW OFTEN?	

HOW OFTEN | MAIN IMPACT

PROBLEM STATEMENT (ONE SENTENCE: WHAT, WHERE, WHEN, HOW MUCH)

STEP 2: MEASURE THE IMPACT

Not sure how to cost it? Use the Kaizen XP KPI Impact Calculator.

DOWNTIME
minutes a week

WASTE
units, kg or packs a week

COST
£ a week, rough is fine

PEOPLE TIME
hours spent dealing with it

SAFETY, QUALITY OR CUSTOMER IMPACT

STEP 3: CONTAIN IT

Stop it hurting today. This buys you time. It is not the fix.

CONTAINMENT ACTION	OWNER	DONE BY	STATUS

STEP 4: BUILD THE TIMELINE

What happened, in order. Patterns hide in the order of events.

DATE / TIME	WHAT HAPPENED	HOW DO I KNOW?

STEP 5: LIST EVERY POSSIBLE CAUSE

Write down everything, even the unlikely ones. Then rule them out with evidence, not a hunch.

PEOPLE — *Training, experience, shift, handover*

MACHINE — *Wear, settings, sensors, maintenance*

METHOD — *Standard work, sequence, changeover*

MATERIAL — *Supplier, batch, spec, storage*

MEASUREMENT — *Gauges, checks, data, calibration*

ENVIRONMENT — *Heat, damp, dust, lighting, layout*

RULE THEM OUT

Every cause gets checked. "Seems unlikely" is not a check.

POSSIBLE CAUSE	HOW I CHECKED IT	RESULT

STEP 6: FIND THE ROOT CAUSE

Take what is left after Step 5 and keep asking why until you hit something you can fix.

STARTING POINT (THE CONFIRMED CAUSE FROM STEP 5)

1 WHY?

2 WHY?

3 WHY?

4 WHY?

5 WHY?

ROOT CAUSE

CATEGORY

PROVEN?

HOW I PROVED IT (THE TEST, THE DATA, OR THE TRIAL)

THE TEST

If you fix this, does the problem stop? If you are not sure, it is not the root cause yet. If the answer is a person, ask why the system let it happen and go again.

STEP 7: FIX IT FOR GOOD

Every action aims at the root cause, has a name next to it, and has a date.

PERMANENT ACTION	OWNER	DUE	STATUS

STEP 8: CHECK, LOCK IT IN, SPREAD IT

HOW WILL WE KNOW IT WORKED? (THE NUMBER WE WILL CHECK)

CHECK DATE | BEFORE | AFTER

RESULT | LOCKED IN BY

HOW IT WAS LOCKED IN (THE DETAIL)

WHERE ELSE COULD THIS HAPPEN? (OTHER LINES, SHIFTS, SITES, MACHINES)

SIGN-OFF

INVESTIGATION LEAD | MANAGER

DATE CLOSED | STATUS

WHO SAW IT?

A finished RCFA nobody senior reads is a favour you did the company. Walk it through with your line leader, and put the result in your Weekly Wins email.

ROOT CAUSE FOUND

Now make it count.

The rest of the Kaizen XP range

THE VISIBILITY KIT

Six tools that make sure the work you already do actually gets seen.

THE FRONTLINE CI STARTER KIT

Five simple tools for spotting problems and making the fix stick.

THE MTBA ASSIST TRACKER

Measure how often a job needs a hand, then prove you fixed it.

THE KPI IMPACT CALCULATOR

Put a pound sign on any problem in two minutes.

THE BREAKDOWN ANALYSIS DASHBOARD

MTBF, MTTR and a Pareto of which machine to fix first.

A closed RCFA is the best entry you will ever put in your Visibility Kit.

All available from Kaizen XP.

XP

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